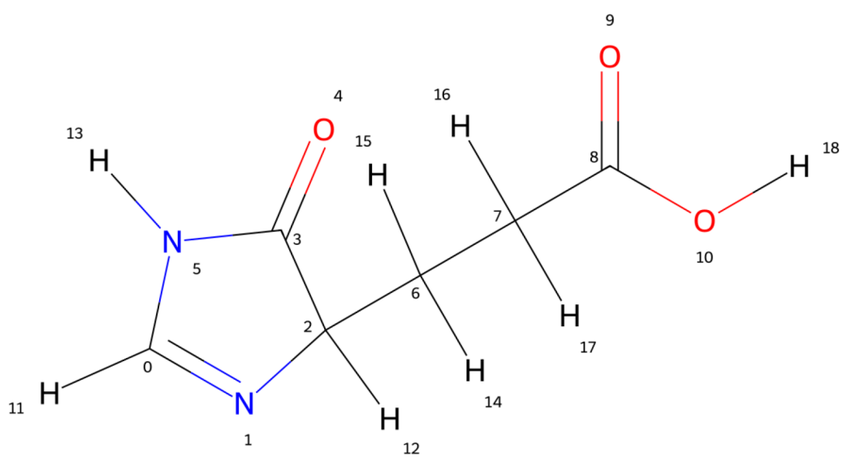
QM Benchmark Report

Potentials: openff-2.3.0, aceff-2.0
Molecules: 2

mol_0 (HEXMLHKQVUFYME-DEPUQRHONA-N)
SMILES: O=C(O)CCC1N=CNC1=O

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Molecule: mol_0 (HEXMLHKQVUFYME-DEPUQRHONA-N), num_conformers: 3

RMSD vs QM (Angstrom)  [mean +/- std across conformers]
-------------------------------------------------------
  openff-2.3.0                    0.1029 +/- 0.0297
  aceff-2.0                       0.0316 +/- 0.0018


ANGLE DIFF vs QM (degrees) > threshold
==============================================================================
Key         QM Ref           openff-2.3.0    aceff-2.0
                             (diff)          (diff)
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(5, 0, 11)  120.057 ± 0.012  6.225 ± 0.014   0.078 ± 0.017
(0, 5, 3)   108.767 ± 0.014  5.718 ± 0.017   0.235 ± 0.009
(1, 0, 5)   115.563 ± 0.056  7.396 ± 0.047   0.166 ± 0.017
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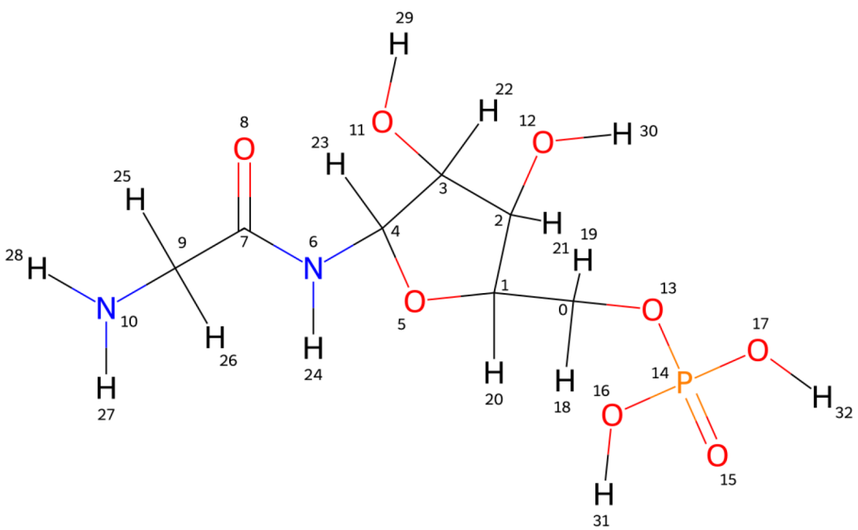
mol_1 (OBQMLSFOUZUIOB-RAIZMQIKNA-N)
SMILES: NCC(=O)NC1OC(COP(=O)(O)O)C(O)C1O

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Molecule: mol_1 (OBQMLSFOUZUIOB-RAIZMQIKNA-N), num_conformers: 3

RMSD vs QM (Angstrom)  [mean +/- std across conformers]
-------------------------------------------------------
  openff-2.3.0                    0.5459 +/- 0.0502
  aceff-2.0                       0.2952 +/- 0.0445


ANGLE DIFF vs QM (degrees) > threshold
==============================================================================
Key           QM Ref           openff-2.3.0    aceff-2.0
                               (diff)          (diff)
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(14, 16, 31)  112.176 ± 1.061  10.043 ± 0.976  0.439 ± 0.033
(14, 17, 32)  111.280 ± 0.979  9.179 ± 0.942   0.147 ± 0.123
(0, 13, 14)   121.816 ± 0.266  5.891 ± 0.332   0.470 ± 0.097
(16, 14, 17)  101.175 ± 0.396  8.018 ± 0.320   1.512 ± 0.275
(13, 14, 16)  102.638 ± 2.407  8.938 ± 4.570   0.312 ± 0.084
(13, 14, 17)  104.321 ± 2.183  5.521 ± 4.403   0.607 ± 0.333
(15, 14, 16)  117.059 ± 1.866  9.548 ± 0.910   1.072 ± 0.511
(15, 14, 17)  115.585 ± 1.857  10.273 ± 0.868  0.385 ± 0.368


TORSION DIFF vs QM (degrees) > threshold
==============================================================================
Key               QM Ref              openff-2.3.0     aceff-2.0
                                      (diff)           (diff)
---------------  ------------------  ---------------  -------------
(0, 13, 14, 15)  -60.746 ± 166.464   60.783 ± 6.578   3.801 ± 1.177
(0, 13, 14, 16)  51.505 ± 2.901      62.863 ± 13.502  4.090 ± 0.754
(0, 13, 14, 17)  -53.623 ± 2.790     67.996 ± 3.499   3.392 ± 1.134
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